Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)	Valid method e.g. $B = \mu_0 nI$ to calculate B **or** I **or** n **or** μ_0 (1) Gives B = 0.0606 [T] **or** I = 7.13 [A] **or** n = 26 972 **or** μ_0 = 2.51 × 10^{-6} (1) Correct conclusion based on calculation (with **ecf**) e.g. Lindsay wrong (if B or I calculated), Lindsay wrong because μ_0 too large or real value is 1.26 × 10^{-6} or Lindsay wrong because n value too large or real value is 13 500 (allow **ecf** on n = 27 000 here) (1) i.e. Calculating B = 0.121 [T] followed by Lindsay correct is 2 marks (1st and 3rd marks)			3	3	1	
		(ii)	Iron / ferromagnetic / steel core Accept increase permeability Don't accept increase permeability of free space	1			1		
	(b)	(i)	Use of $BANcos\theta$ (1) Correct answer = 0.046 Wb **or** Wb turn **or** T m^2 ****unit mark**** (1)	1	1		2	1	
		(ii)	Max when $\theta = 90$ because flux changing / cutting at max rate (accept cutting most lines or similar for max rate) (1) When $\theta = 0$ no emf because flux max or no cutting or rate of change is zero (1) **Alternative for 2nd mark** Min when $\theta = -90, 270$ because flux changing / cutting in opposite direction (1) Allow 1 mark if **both** angles are correct with no explanation Allow 1 mark if angles are wrong but **both** explanations are correct (must refer to flux **changing** or **cutting** but no need to insist on "rate")		2		2		
			Question 1 total	**2**	**3**	**3**	**8**	**2**	**0**